

Lab 6: MOSFETs

ECE 0102: Microelectronic Circuits + Lab

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Team Roles:

Activity	Student Name
Circuit Construction	Stefan Zhang
Data Collection	Knight Liu
Data Analysis	Stefan Zhang; Knight Liu
Answers to Questions	Stefan Zhang; Knight Liu
Lab Report Writing	Stefan Zhang

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1) Objectives and Background

The objective of this lab is to measure the I_D - V_{DS} characteristics of an NMOS transistor and to extract a simple MOSFET model using the threshold voltage V_{tn} and the transconductance parameter k_n [1]. The NMOS device was taken from a CD4007UB CMOS transistor array, using pin 6 as the gate, pin 7 as the source, and pin 8 as the drain [3]. During the experiment, V_{GS} was fixed at 3 V and V_{DS} was varied from 0.5 V to 5.5 V.

2) Equipment Used

Table 1: Equipment used in Lab 6

Item	Quantity
CD4007UB CMOS transistor array	1
Resistor, approximately 1 k Ω	1
DC Power Supply	1
Digital Multimeter	1
Breadboard and wires	1 set

3) Theory

3.1) NMOS Operation Regions

For an enhancement-type NMOS transistor, the transistor is turned on when the gate-source voltage is larger than the threshold voltage. The following operation-region equations are based on the standard long-channel NMOS model [2]:

$$V_{GS} > V_{tn}$$

The overdrive voltage is defined as

$$V_{OV} = V_{GS} - V_{tn}$$

The operation region is determined by comparing V_{DS} with V_{OV} . If

$$V_{DS} < V_{OV}$$

the transistor operates in the triode region. The drain current is modeled as

$$I_D = k_n \left[(V_{GS} - V_{tn})V_{DS} - \frac{1}{2}V_{DS}^2 \right]$$

If

$$V_{DS} \geq V_{OV}$$

the transistor operates in the saturation region. The drain current is modeled as

$$I_D = \frac{1}{2}k_n(V_{GS} - V_{tn})^2$$

In this lab, channel-length modulation is ignored according to the Lab 6 instruction [1]. Therefore, the factor $(1 + \lambda V_{DS})$ is not included.

3.2) Current Measurement Method

The drain current can be obtained from the voltage drop across the drain resistor, following the measurement method specified in the lab manual [1]:

$$I_{D1} = \frac{V_{DD} - V_{DS}}{R_D}$$

where R_D is the measured drain resistor value. In this experiment,

$$R_D = 992.5 \, \Omega$$

The measured drain current I_{D1} is compared with the calculated model current I_{D2} .

The percentage error is calculated by

$$\text{Error} = \left| \frac{I_{D2} - I_{D1}}{I_{D1}} \right| \times 100\%$$

4) Experimental Setup

The source terminal of the NMOS transistor was connected to ground. The gate terminal was connected to a fixed 3 V DC supply. The drain terminal was connected to the bottom of the drain resistor R_D , while the top of the resistor was connected to the adjustable DC supply V_{DD} .

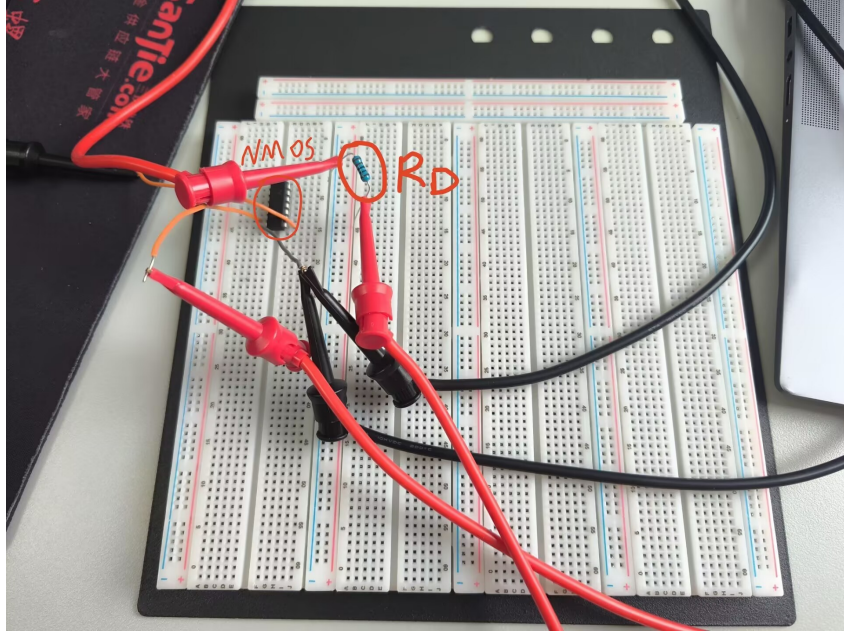


Figure 1: Experimental setup for measuring the NMOS I-V characteristics

5) Experimental Data

The measured value of the drain resistor was

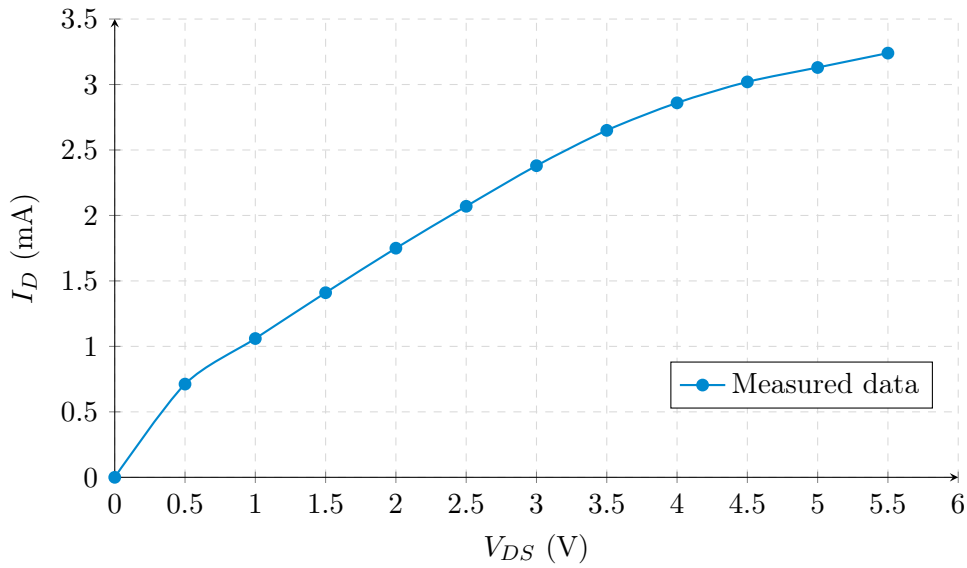
$$R_D = 992.5 \, \Omega$$

The measured current values are shown in Table 2. The measured current I_{D1} was taken from the recorded experimental data.

Table 2: Measured and calculated NMOS drain current

V_{DD} [V]	V_{DS} [V]	I_D from Action #1	I_D from Action #2	Error [%]
1.0	0.5	7.12×10^{-4}	7.12×10^{-4}	0.00
1.5	1.0	1.06×10^{-3}	1.06×10^{-3}	0.00
2.0	1.5	1.41×10^{-3}	1.10×10^{-3}	22.14
2.5	2.0	1.75×10^{-3}	1.10×10^{-3}	37.27
3.0	2.5	2.07×10^{-3}	1.10×10^{-3}	46.96
3.5	3.0	2.38×10^{-3}	1.10×10^{-3}	53.87
4.0	3.5	2.65×10^{-3}	1.10×10^{-3}	58.57
4.5	4.0	2.86×10^{-3}	1.10×10^{-3}	61.61
5.0	4.5	3.02×10^{-3}	1.10×10^{-3}	63.65
5.5	5.0	3.13×10^{-3}	1.10×10^{-3}	64.92
6.0	5.5	3.24×10^{-3}	1.10×10^{-3}	66.05
Average Error				43.19

The measured I_D - V_{DS} curve is shown in Fig. 2.

**Figure 2:** Measured I_D - V_{DS} characteristic of the NMOS transistor for $V_{GS} = 3$ V.

6) Model Extraction

6.1) Determination of V_{tn} and k_n

The model was developed using the standard long-channel NMOS equations [2]. Since the lab manual suggested that the threshold voltage should be around 1 V to 2 V, with a target near 1.8 V [1], the low- V_{DS} data points were used to estimate the model parameters.

For the first two measured points, the transistor is treated as operating in the triode region. Therefore,

$$I_D = k_n \left[(V_{GS} - V_{tn})V_{DS} - \frac{1}{2}V_{DS}^2 \right]$$

Using the first two data points,

$$V_{DS} = 0.5 \text{ V}, \quad I_D = 7.12 \times 10^{-4} \text{ A}$$

and

$$V_{DS} = 1.0 \text{ V}, \quad I_D = 1.06 \times 10^{-3} \text{ A}$$

the extracted threshold voltage and transconductance parameter are

$$V_{tn} = 1.77 \text{ V}$$

and

$$k_n = 1.456 \times 10^{-3} \text{ A/V}^2$$

Thus,

$$V_{OV} = V_{GS} - V_{tn} = 3.00 - 1.77 = 1.23 \text{ V}$$

For $V_{DS} < 1.23 \text{ V}$, the triode-region equation was used. For $V_{DS} \geq 1.23 \text{ V}$, the saturation-region equation was used.

6.2) Extracted Model Parameters

Table 3: Extracted NMOS model parameters

Parameter	Value
$R_D \text{ } (\Omega)$	992.5
$V_{tn} \text{ (V)}$	1.77
$k_n \text{ (A/V}^2\text{)}$	1.456×10^{-3}

7) Discussion

The extracted threshold voltage was $V_{tn} = 1.77\text{ V}$, close to the expected value of about 1.8 V . With $V_{GS} = 3\text{ V}$, the overdrive voltage was $V_{OV} = 1.23\text{ V}$, so the theoretical transition from triode to saturation should occur near $V_{DS} = 1.23\text{ V}$.

The measured data do not show an ideal flat saturation region. Instead, I_D continues to increase as V_{DS} increases. This explains the large error at high V_{DS} values in Table 2. A likely reason is channel-length modulation: in a real MOSFET, increasing V_{DS} shortens the effective channel length and increases the saturation current [2]. Since this effect was intentionally ignored in the required model [1], the simplified model matches the low- V_{DS} data better than the high- V_{DS} data.

8) Conclusion

The I_D - V_{DS} characteristics of an NMOS transistor from a CD4007UB chip were measured at $V_{GS} = 3\text{ V}$. From the low- V_{DS} data, the extracted model parameters were

$$V_{tn} = 1.77\text{ V}, \quad k_n = 1.456 \times 10^{-3}\text{ A/V}^2$$

The simplified MOSFET model describes the low- V_{DS} behavior reasonably well, but it gives larger error at high V_{DS} because channel-length modulation was not included.

References

- [1] SCUPI 2026S Microelectronics Circuit Lab Manual, Lab #6: MOSFETs.
- [2] Sedra, A. S., Smith, K. C., *Microelectronic Circuits*, Oxford University Press.
- [3] Texas Instruments, *CD4007UB CMOS Dual Complementary Pair Plus Inverter Datasheet*.